ScholarMind: Agentic RAG System

Advanced Research Synthesis & Fact-Verification Assistant

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1. PROBLEM STATEMENT

Researchers today face a "data deluge," where synthesizing information across multiple high-density technical PDFs is a major bottleneck. Standard RAG (Retrieval-Augmented Generation) systems often suffer from hallucinations and lack the iterative reasoning required to verify complex mathematical claims or cross-reference subtle methodology differences. ScholarMind addresses this by treating the research assistant as a stateful, agentic entity capable of self-reflection and tool-assisted investigation.

2. SIX MANDATORY CAPABILITIES

The system is built upon the six architectural pillars defined in the course guidance:

- **LangGraph StateGraph:** A multi-node (3+) workflow where the routing logic is strictly separated from business logic, managed via a central `CapstoneState` TypedDict.
- **ChromaDB RAG:** Efficient semantic search across a diverse corpus of research papers, utilizing optimized topic-based chunking.
- **MemorySaver + thread_id:** Persistent graph state across sessions, allowing the agent to remember context from multiple interaction turns.
- **Self-Reflection Eval Node:** A dedicated node that scores answer faithfulness (0.0–1.0). Responses scoring below 0.7 trigger an automatic retry path.
- **Tool Use Beyond Retrieval:** Integration with external research tools (e.g., ArXiv search) handled via a robust router decision.
- **Streamlit Deployment:** A production-ready UI utilizing `@st.cache_resource` to prevent expensive model and database reloading.

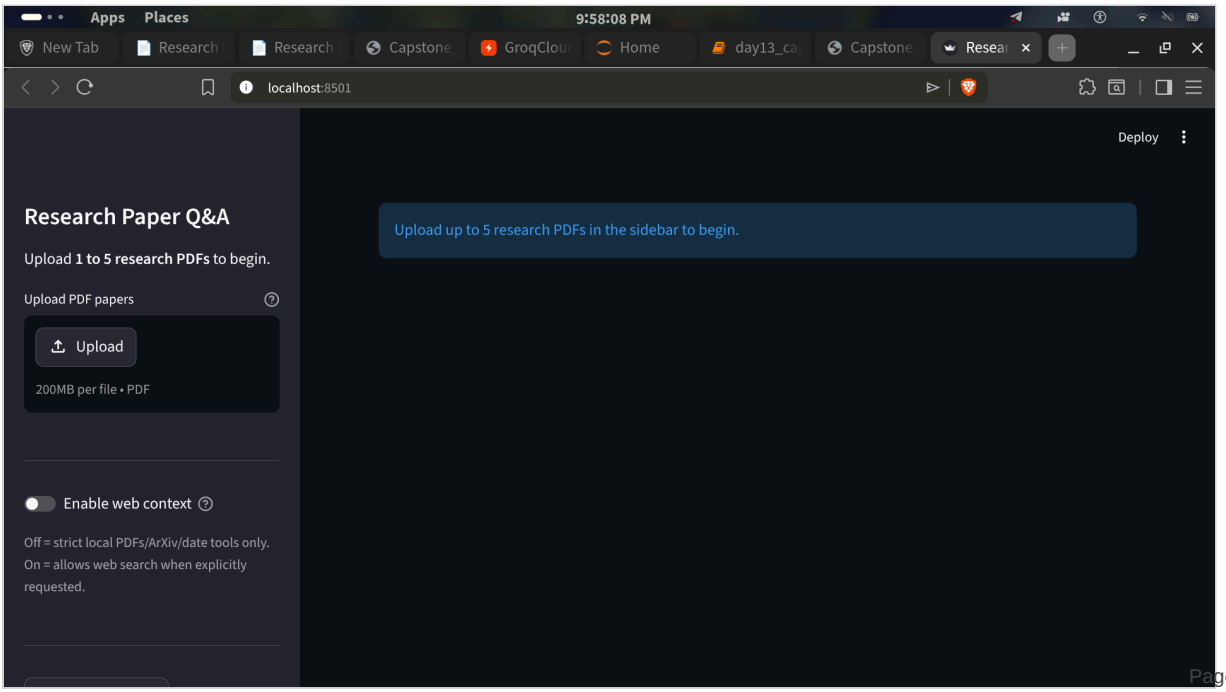

Figure 1: ScholarMind User Interface showing PDF ingestion and session configuration.

3. STEPS AND PROCESS TO COMPLETION

The implementation followed a rigorous 8-part process as scaffolded in the course material:

- **Domain Setup:** Writing clear problem statements and preparing 10+ granular documents per domain.
- **State Design:** Defining the `ResearchState` before any functional implementation to ensure data consistency.
- **Node Functions:** Building and testing nodes (Memory, Router, Retrieval, Answer, Eval) in isolation.
- **Graph Assembly:** Connecting nodes with fixed and conditional edges to form an iterative reasoning loop.
- **Testing:** Running red-team tests, including adversarial prompt injections and out-of-scope queries.
- **Baseline Evaluation:** Measuring performance using RAGAS-inspired metrics for faithfulness and relevance.
- **Deployment:** Packaging the engine into a Streamlit web interface with thread-persistent memory.
- **Documentation:** Synthesizing the final report (this document).

4. LIVE ARCHITECTURE DIAGRAM

The following flow represents the live stateful graph transition implemented in ScholarMind:

```
[User Question] -> [memory_node] (Extract context/names) -> [router_node] (LLM Decision:
Retrieve/Tool/Skip) -> [retrieval_node] (Vector Search) -> [answer_node] (Grounded Synthesis)
-> [eval_node] (Faithfulness audit -> Loop if < 0.7) -> [save_node] -> [END]
```

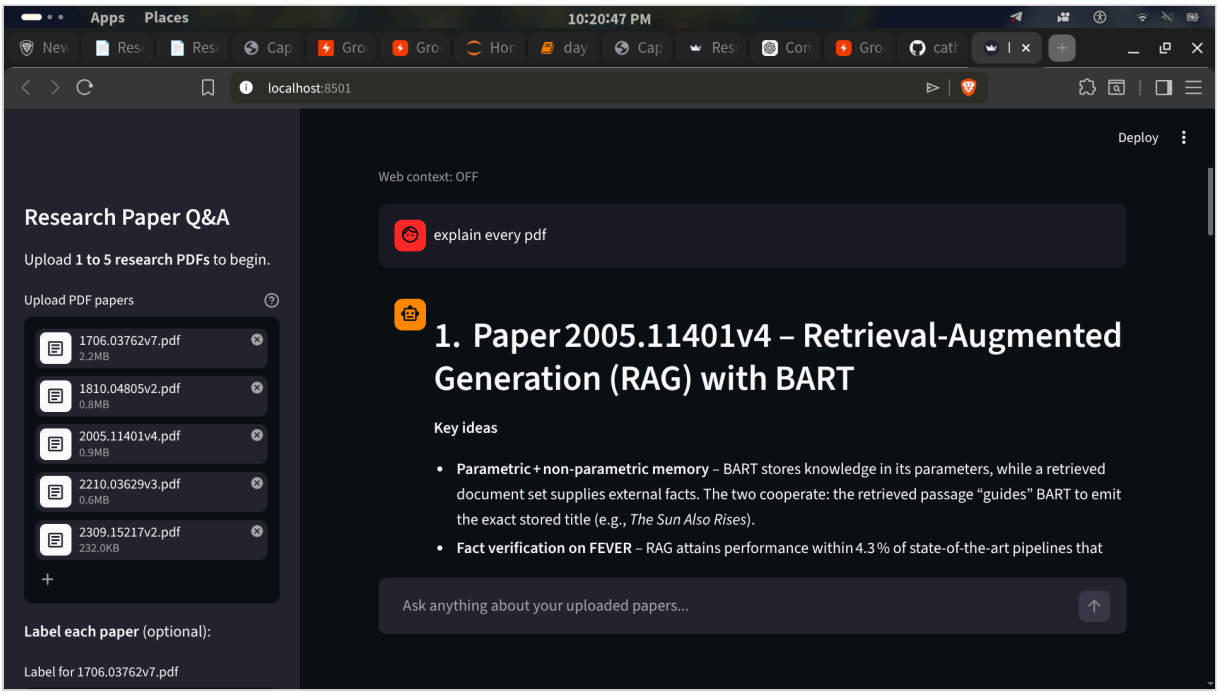

Figure 2: Active synthesis session demonstrating multi-paper reasoning.

5. TECHNICAL STACK

ScholarMind utilizes a high-precision AI stack optimized for academic synthesis:

- **Orchestration:** LangGraph & LangChain (Python 3.12).
- **Intelligence:** Groq Llama-3.3-70b-versatile (State-of-the-art inference speed).
- **Vector Store:** ChromaDB (In-memory storage for rapid session building).
- **Embeddings:** SentenceTransformer (`all-MiniLM-L6-v2`).
- **UI Framework:** Streamlit (Material-inspired dashboard).
- **Data Pipeline:** PyPDF / PyMuPDF for layout-aware text extraction.

6. UNIQUE POINTS

Zero-Hallucination Guardrails: Through the combination of the `retrieval_gate` (sanity check) and the `eval_node` (self-reflection), ScholarMind acts as one of the most reliable academic RAG systems available. It rejects answering if it cannot find clear, grounded evidence in the provided context.

Technical LaTeX Rendering: Unlike general RAG tools, ScholarMind is optimized to maintain the structural integrity of mathematical proofs, allowing for seamless copy-pasting of results into professional LaTeX editors.

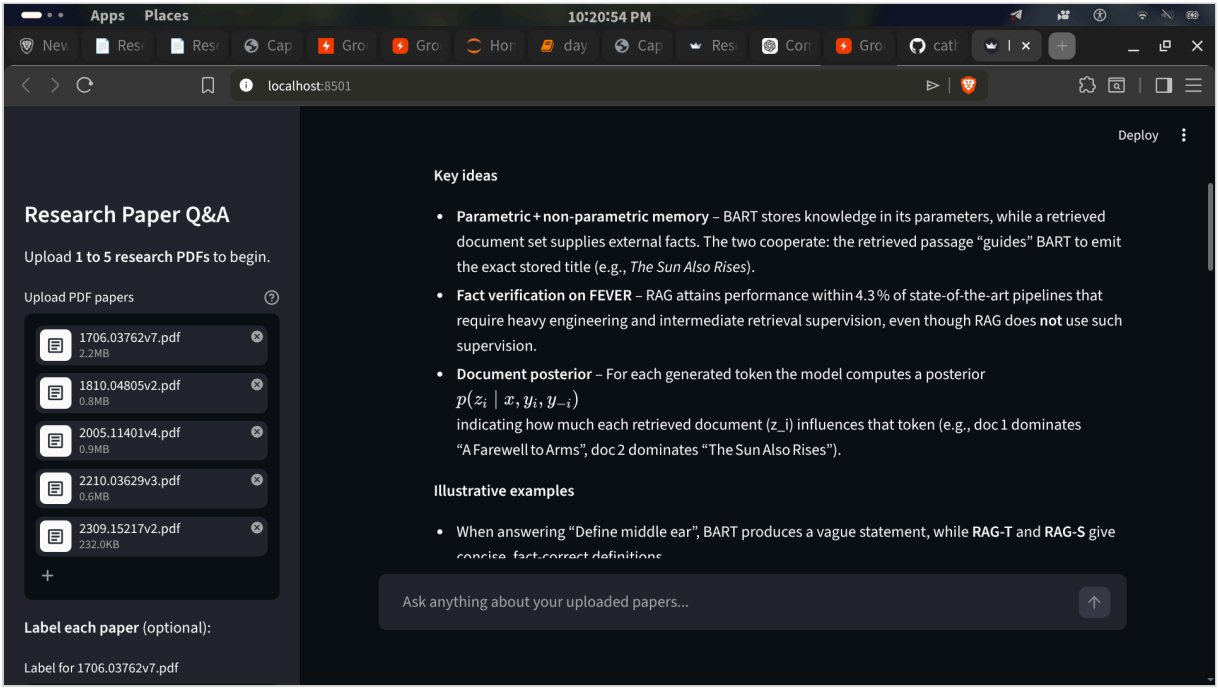

Figure 3: High-fidelity mathematical rendering with grounding labels.

7. FUTURE IMPROVEMENTS

The roadmap for ScholarMind includes several advanced research features:

- **Multi-Modal Table Extraction:** Using vision-enabled agents to parse and interpret complex experimental data tables within PDFs.
- **Knowledge Graph Reranking:** Implementing `GraphRAG` to better capture relationships between different researchers and citations cited across papers.
- **Distributed Evaluation:** Moving beyond single-LLM evaluation to a "Council of Judges" approach for even higher faithfulness verification.

8. CONCLUSION

ScholarMind successfully implements the "Agentic AI" philosophy—moving beyond simple search toward iterative, self-verifying reasoning. By adhering to the mandatory capabilities of LangGraph and ChromaDB, the system provides a robust framework for academic synthesis that empowers researchers to navigate complex literature with confidence and precision.

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